Lembit Mihkelstein

Senior AI/ML Software Engineer

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Profile

Senior AI Architect and Machine Learning Engineer with 10+ years of experience building production grade AI systems from classical ML forecasting to large scale LLM orchestration and agent driven platforms. Proven track record designing end to end AI lifecycles including data engineering, model training, reinforcement learning, MLOps automation, and real time application delivery across Google Cloud and AWS. Strong expertise in scalable backend systems, distributed data pipelines, and low latency AI powered user interfaces serving enterprise and real world industrial use cases, delivering measurable ROI, innovation, reliability, and strategic transformation.

Professional Experience

Senior AI/ML Software Engineer

Nov 2023 - May 2026

Indeema | Kraków, Poland

- Architected and deployed text-to-application AI platform using Python 3.11, FastAPI 0.110, and Google Cloud Platform, generating production-ready mini applications in real time.
- Implemented LLM orchestration pipelines with Vertex AI 1.47, OpenAI GPT-4o, and Anthropic Claude 3, enabling structured tool calling for automated code and logic generation.
- Developed machine learning–based intent routing using Hugging Face Transformers 4.38, PyTorch 2.2, and scikit-learn 1.4, producing structured execution outputs.
- Designed scalable data processing workflows with Apache Beam 2.54, BigQuery ML, and Cloud Dataflow, standardizing multi-source datasets for AI consumption.
- Built backend microservices on Cloud Run with Firestore and Cloud SQL (PostgreSQL 15), ensuring isolated business logic and persistent storage per generated application.
- Automated end-to-end MLOps lifecycle using Docker 25, Cloud Build, and Vertex AI Pipelines, enabling CI/CD-driven training and deployment.
- Developed real-time frontend layer with React 18, Next.js 14, and TypeScript 5, integrated with Pub/Sub streaming and Memorystore (Redis 7) for low-latency dashboards.

AI/ML Software Engineer

May 2020 - Oct 2023

Future Processing | Gliwice, Poland

- Delivered production AI headshot lighting correction system using Python 3.8, Django 3.0, and AWS EC2, exposing scalable inference REST APIs.
- Built synthetic data generation pipeline with TensorFlow 2.6, Keras 2.6, and AWS SageMaker, producing identity-diverse training datasets with controlled illumination.
- Collected paired real-world lighting datasets via automated photobooth system using Arduino Mega 2560, gPhoto2 2.5, and AWS S3, generating multi-exposure ground truth data.
- Trained supervised lighting normalization models with PyTorch 1.10, Albumentations 1.1, and scikit-learn 1.0, producing validated deployable artifacts.
- Orchestrated ETL and dataset pipelines with Apache Airflow 2.2, AWS Glue, and AWS Athena, enabling reproducible and structured data workflows.
- Implemented containerized microservices using Docker 20.10, AWS ECS Fargate, PostgreSQL 13, and Redis 6 for low-latency inference and metadata caching.
- Established MLOps and monitoring stack with GitHub Actions, Terraform 1.2, MLflow 1.30, Prometheus 2.37, Grafana 8.5, and React 17, supporting real-time updates via AWS API Gateway (WebSocket) and global CDN delivery through AWS CloudFront.

Education

Master's Degree in Computer Science

University of Tartu

Sep 2010 - Jun 2015

Competencies

- AI Architecture and System Design
- ML and DL
- LLM and Agentic Workflows
- Reinforcement Learning
- Data Engineering and ETL Pipelines
- MLOps and CI/CD Automation
- Cloud Native Backend Systems
- Real Time AI Applications

Technical Skills

Programming Languages

Python, JavaScript, TypeScript, SQL, Bash

AI and LLM Systems

Vertex AI, OpenAI GPT 4o, LangChain, Anthropic Claude 3, RAG systems, LLM tool calling pipelines,
structured output generation,
prompt engineering,
agent orchestration

ML and DL

PyTorch, TensorFlow, Keras, scikit learn, XGBoost, LightGBM, Hugging Face Transformers, Reinforcement Learning, OpenAI Gym, Proximal Policy Optimization

Cloud Platform

Google Cloud Platform, AWS EC2, AWS SageMaker, BigQuery, Cloud SQL, S3, Pub Sub, Memorystore, CloudFront

Artificial Intelligence Engineer
Sep 2018 - Apr 2020

DATAFOREST | Tallinn, Estonia

- Designed reinforcement learning control system for refinery workflow automation using Python 3.8, TensorFlow 2.1, and OpenAI Gym 0.17, delivering dynamic CDU optimization policies.
- Implemented multi agent architecture and reward modeling with Keras 2.3, NumPy 1.18, and SciPy 1.4, enabling adaptive decision making under operational constraints.
- Built high fidelity refinery simulation environment using Pandas 1.0, SimPy 3.0, and Matplotlib 3.2, generating large scale synthetic experience data for safe RL training.
- Developed policy optimization and training pipelines with Proximal Policy Optimization, TensorBoard 2.1, and HDF5, producing stable convergence across thousands of simulated refinery states.
- Engineered experiment tracking and reproducible model lifecycle workflows using MLflow 1.8, Docker 19.03, and Git 2.25, ensuring controlled training iterations and model versioning.

Machine Learning Engineer
Jul 2015 - Aug 2018

MindTitan | Tallinn, Estonia

- Developed yield and margin prediction models using Python 3.6, scikit-learn 0.19, XGBoost 0.71, and LightGBM 2.1, delivering field level production forecasts.
- Built CNN and LSTM models for crop classification and seasonal trend modeling with TensorFlow 1.10, Keras 2.2, and PyTorch 0.4, leveraging multispectral imagery.
- Implemented time series forecasting pipelines using Prophet 0.3, statsmodels 0.9, and custom LSTM architectures for weather driven output prediction.
- Engineered agronomic and geospatial features using Pandas 0.23, NumPy 1.15, and GeoPandas 0.4, producing structured model ready datasets.
- Performed hyperparameter tuning and validation with GridSearchCV, RandomizedSearchCV, and Optuna 0.9, improving cross region generalization.
- Ensured experiment tracking and reproducibility via MLflow 0.9, DVC 0.30, and structured evaluation using SHAP 0.28 and cross validation metrics.

Projects

AI-Powered Precision Agriculture & Yield Intelligence Platform
https://granular.ai

Led the development of an AI driven precision agriculture platform that unified agronomic, satellite, weather, and financial data to deliver scalable crop yield forecasting, field level profitability prediction, and risk aware decision support for enterprise farming operations.

AI-Powered Text to App Platform
https://moneta.studio

Designed and delivered a scalable AI powered text to app platform that transforms natural language prompts into fully functional, deployable, and collaborative mini applications with real time data processing and dynamic user interaction.

Technical Skills

Data Engineering & Processing
Apache Beam, Apache Airflow, BigQuery ML, Cloud Dataflow, AWS Glue, AWS Athena, Pandas, NumPy, GeoPandas, feature engineering, large scale ETL pipelines

Backend & Microservices
FastAPI, Django, RESTful API, WebSockets Cloud Run, AWS ECS Fargate, Redis, API Gateway, Firestore, PostgreSQL, microservices architecture

MLOps & DevOps
Docker, Kubernetes, Terraform, MLflow, Vertex AI Pipelines, Cloud Build, GitHub Actions, CI/CD automation, model versioning, experiment tracking, monitoring with Prometheus and Grafana

Frontend & Real-Time Systems
React, Next.js, TypeScript, streaming dashboards, low latency data visualization, event driven architectures

Soft Skills
Strong Communication, Cross- Functional Team Collaboration, Leadership, Stakeholder Communication, Mentorship, Adaptability, Creativity & Risk-Taking, Problem Solving, Team Building, Conflict Resolution, Problem-Solving, Agile & Scrum Methodologies, Time Management, Business Needs Understanding